CHAPTER 4: ELECTROMAGNETIC INDUCTION

1. Faraday's Law of Induction:
The induced electromotive force (EMF) in any closed circuit is equal to the negative of the time rate of change of magnetic flux through the circuit.
Formula: EMF = -d(Phi_B)/dt

2. Lenz's Law:
The direction of the induced current is such that it opposes the change in magnetic flux that produced it, ensuring conservation of energy.

3. Mutual and Self Inductance:
Self-inductance L = (N * Phi) / I. Unit: Henry (H).